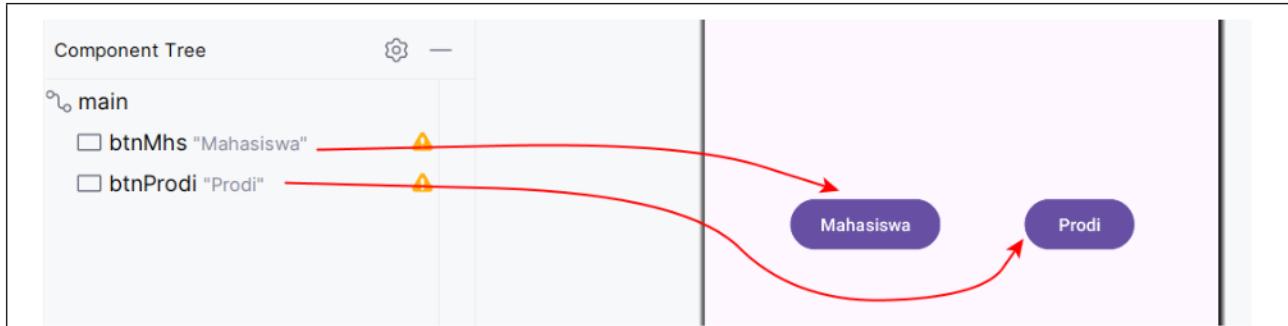
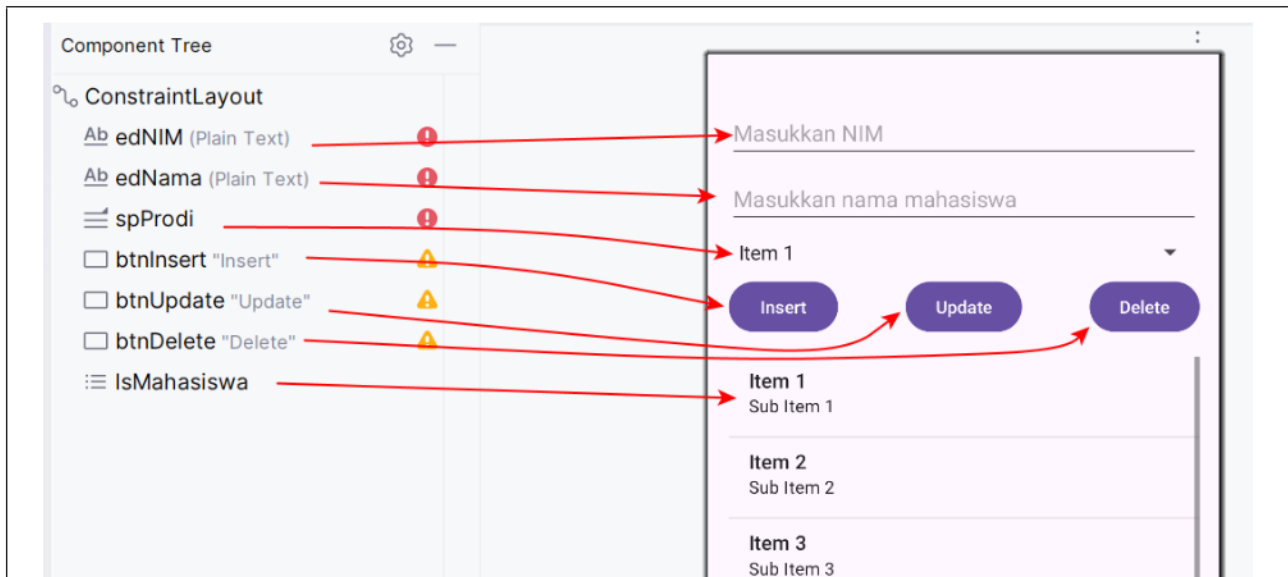


SQLite Database

A. UI activity_main.xml



B. UI activity_mahasiswa.xml



C. UI item_data_mahasiswa.xml



D. build.gradle.kts (Module: app)

```
plugins {
    alias(libs.plugins.android.application)
```

```
}  
  
android {  
    namespace = "nugroho.benni.mobile08c"  
    compileSdk {  
        version = release(36) {  
            minorApiLevel = 1  
        }  
    }  
  
    defaultConfig {  
        applicationId = "nugroho.benni.mobile08c"  
        minSdk = 26  
        targetSdk = 36  
        versionCode = 1  
        versionName = "1.0"  
  
        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"  
    }  
  
    buildFeatures {  
        viewBinding = true  
    }  
  
    buildTypes {  
        release {  
            isMinifyEnabled = false  
            proguardFiles(  
                getDefaultProguardFile("proguard-android-optimize.txt"),  
                "proguard-rules.pro"  
            )  
        }  
    }  
    compileOptions {  
        sourceCompatibility = JavaVersion.VERSION_11  
        targetCompatibility = JavaVersion.VERSION_11  
    }  
}  
  
dependencies {  
    implementation(libs.androidx.core.ktx)  
    implementation(libs.androidx.appcompat)  
    implementation(libs.material)  
    implementation(libs.androidx.activity)  
    implementation(libs.androidx.constraintlayout)  
    testImplementation(libs.junit)  
    androidTestImplementation(libs.androidx.junit)  
    androidTestImplementation(libs.androidx.espresso.core)  
}
```

E. Class DBOpenHelper.kt

```
package nugroho.benni.mobile08c

import android.content.Context
import android.database.sqlite.SQLiteDatabase
import android.database.sqlite.SQLiteOpenHelper

class DBOpenHelper(context : Context) :
    SQLiteOpenHelper(context, name, null, version) {
    override fun onCreate(db: SQLiteDatabase?) {
        val sqlProdi = "create table prodi(" +
            "id_prodi integer primary key, " +
            "nama_prodi text not null)"
        val sqlMahasiswa = "create table mahasiswa(" +
            "nim text primary key," +
            "nama_mahasiswa text not null," +
            "id_prodi integer not null)"
        val insertProdi = "insert into prodi(nama_prodi) " +
            "values " +
            "('D3 Manajemen Informatika')," +
            "('D3 Teknik Mesin')," +
            "('D3 Akuntansi)"
        db?.execSQL(sqlProdi)
        db?.execSQL(sqlMahasiswa)
        db?.execSQL(insertProdi)
    }

    override fun onUpgrade(db: SQLiteDatabase?, p1: Int, p2: Int) {
        TODO("Not yet implemented")
    }

    companion object {
        private const val name = "SIAKAD.db"
        private const val version = 1
    }
}
```

F. Class MainActivity.kt

```
package nugroho.benni.mobile08c

import android.content.Intent
import android.os.Bundle
import android.view.View
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import nugroho.benni.mobile08c.databinding.ActivityMainBinding

class MainActivity : AppCompatActivity(), View.OnClickListener {
    lateinit var b : ActivityMainBinding
}
```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    b = ActivityMainBinding.inflate(layoutInflater)
    enableEdgeToEdge()
    setContentView(b.root)
    ViewCompat.setOnApplyWindowInsetsListener(
        findViewById(R.id.main)) { v, insets ->
            val systemBars =
                insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
                systemBars.bottom)
            insets
        }
    b.btnMhs.setOnClickListener(this)
    b.btnProdi.setOnClickListener(this)
}

override fun onClick(p0: View?) {
    when(p0?.id){
        R.id.btnMhs -> {
            val intent = Intent(this, MahasiswaActivity::class.java)
            startActivity(intent)
        }
        R.id.btnProdi ->{
            TODO ("Lengkapi kode untuk membuka activity prodi")
        }
    }
}
}

```

G. Class MahasiswaActivity.kt

```

package nugroho.benni.mobile08c

import android.database.Cursor
import android.database.sqlite.SQLiteDatabase
import android.os.Bundle
import android.view.View
import android.widget.AdapterView
import android.widget.AdapterView.OnItemClickListener
import android.widget.CursorAdapter
import android.widget.ListAdapter
import android.widget.SimpleCursorAdapter
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import nugroho.benni.mobile08c.databinding.ActivityMahasiswaBinding

class MahasiswaActivity : AppCompatActivity(), View.OnClickListener,
    AdapterView.OnItemClickListener, AdapterView.OnItemClickListener {
    lateinit var b : ActivityMahasiswaBinding
    lateinit var db : SQLiteDatabase
    lateinit var simpleCursor : SimpleCursorAdapter
    lateinit var listAdapter : ListAdapter
    var namaProdi = ""
    var idProdi = 0
}

```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    b = ActivityMahasiswaBinding.inflate(layoutInflater)
    setContentView(b.root)
    db = DBOpenHelper(this).writableDatabase

    b.btnInsert.setOnClickListener(this)
    b.btnDelete.setOnClickListener(this)
    b.btnUpdate.setOnClickListener(this)
    b.spProdi.onItemSelectedListener = this
    b.lsMahasiswa.onItemClickListener = this
}

override fun onClick(p0: View?) {
    when(p0?.id){
        R.id.btnInsert ->{
            val sql =
                "select id_prodi from prodi where nama_prodi = '$namaProdi'"
            val c = db.rawQuery(sql,null)
            c.moveToFirst()
            val sqlInsert =
                "insert into mahasiswa(nim, nama_mahasiswa, id_prodi) "+
                "values(?,?,?)"
            db.execSQL(sqlInsert, arrayOf(
                b.edNIM.text.toString(), b.edNama.text.toString(),
                c.getString(0)))
            showDataMahasiswa()
        }
        R.id.btnUpdate ->{
            val sql =
                "select id_prodi from prodi where nama_prodi = '$namaProdi'"
            val c = db.rawQuery(sql,null)
            c.moveToFirst()
            val sqlUpdate = "update mahasiswa set " +
                "nama_mahasiswa='${b.edNama.text}', " +
                "id_prodi=${c.getString(0)} " +
                "where nim='${b.edNIM.text}'"
            db.execSQL(sqlUpdate)
            showDataMahasiswa()
        }
        R.id.btnDelete ->{
            val sql = "delete from mahasiswa where nim='${b.edNIM.text}'"
            db.execSQL(sql)
            showDataMahasiswa()
        }
    }
}

override fun onItemSelected(p0: AdapterView<*>?, p1: View?, p2: Int,
    p3: Long
) {
    val c = b.spProdi.getItemAtPosition(p2) as Cursor
    namaProdi = c.getString(c.getColumnIndexOrThrow("_id"))
}

override fun onNothingSelected(p0: AdapterView<*>?) {

```

```
        TODO("Not yet implemented")
    }

    override fun onStart() {
        super.onStart()
        showDataProdi()
        showDataMahasiswa()
    }

    fun showDataProdi(){
        val c = db.rawQuery(
            "select nama_prodi as _id from prodi order by nama_prodi asc",
            null)
        simpleCursor = SimpleCursorAdapter(this,
            android.R.layout.simple_spinner_item, c,
            arrayOf("_id"),
            intArrayOf(android.R.id.text1),
            CursorAdapter.FLAG_REGISTER_CONTENT_OBSERVER)
        simpleCursor.setDropDownViewResource(
            android.R.layout.simple_spinner_dropdown_item)
        b.spProdi.adapter = simpleCursor
        b.spProdi.setSelection(0)
    }

    fun showDataMahasiswa(){
        val c = db.rawQuery(
            "select m.nim as _id, m.nama_mahasiswa, p.nama_prodi " +
            "from mahasiswa m, prodi p " +
            "where m.id_prodi = p.id_prodi " +
            "order by m.nama_mahasiswa asc", null)
        listAdapter = SimpleCursorAdapter(this,
            R.layout.item_data_mahasiswa, c,
            arrayOf("_id", "nama_mahasiswa", "nama_prodi"),
            intArrayOf(R.id.txNIM, R.id.txNama, R.id.txProdi),
            CursorAdapter.FLAG_REGISTER_CONTENT_OBSERVER)
        b.lsMahasiswa.adapter = listAdapter
    }

    override fun onItemClick(p0: AdapterView<*>?, p1: View?, p2: Int, p3: Long
    ){
        val c = p0?.adapter?.getItem(p2) as Cursor
        b.edNIM.setText(c.getString(c.getColumnIndexOrThrow("_id")))
        b.edNama.setText(c.getString(c.getColumnIndexOrThrow("nama_mahasiswa")))
        namaProdi = c.getString(c.getColumnIndexOrThrow("nama_prodi"))
        val c2 = db.rawQuery(
            "select id_prodi, nama_prodi from prodi order by nama_prodi asc",
            null)
        c2.moveToPosition(-1)
        var index = 0
        while (c2.moveToNext()){
            if (c2.getString(1).equals(namaProdi)){
                b.spProdi.setSelection(index)
                idProdi = c2.getInt(0)
            }else{ index++ }
        }
    }
}
```

```
}  
}
```

H. AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<manifest xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:tools="http://schemas.android.com/tools">  
  
    <application  
        android:allowBackup="true"  
        android:dataExtractionRules="@xml/data_extraction_rules"  
        android:fullBackupContent="@xml/backup_rules"  
        android:icon="@mipmap/ic_launcher"  
        android:label="@string/app_name"  
        android:roundIcon="@mipmap/ic_launcher_round"  
        android:supportsRtl="true"  
        android:theme="@style/Theme.Mobile08c">  
        <activity  
            android:name=".MainActivity"  
            android:exported="true">  
            <intent-filter>  
                <action android:name="android.intent.action.MAIN" />  
  
                <category android:name="android.intent.category.LAUNCHER" />  
            </intent-filter>  
        </activity>  
        <activity android:name=".MahasiswaActivity"></activity>  
    </application>  
</manifest>
```

I. Tugas

1. Tambahkan antarmuka dan class untuk melakukan operasi CRUD pada tabel prodi